

CHAPTER FOUR

SYSTEM IMPLEMENTATION AND RESULTS

4.0 Introduction

This chapter explains how the design presented in Chapter Three was implemented as a working personality prediction application and evaluates the resulting eight model pipelines. Python, Django, SQLite, BERT, scikit-learn and PyTorch support the web interface, data storage and machine learning workflow. The chapter describes the development environment, interacting subsystems, dataset and user interfaces, and compares a saved training session with the latest eight successful prediction batches. Continuous OCEAN prediction is the primary evaluation task; threshold-based Low and High classification provides a secondary view of performance.

4.1 Choice of Development Environment and System Requirements

The application uses a Python server with Django views and templates. This environment brings data processing, model execution and result presentation into one application while keeping the machine learning components in separate service modules. HTML, CSS and JavaScript provide the browser interface. The local development database is SQLite, and model weights, embeddings and detailed reports are retained in separate artifact and cache folders. The local research workflow therefore does not require a separate database server.

Table 4.1: Software components and their implementation roles

Component	Specification and role
Programming languages	Python 3.11 in the Dockerfile for server and models; HTML, CSS and JavaScript for interfaces
Web framework	Django 5.0.6 in requirements.txt; server-rendered templates and authentication
Database	SQLite for the local db.sqlite3; relational experiment and prediction records
Representation	Hugging Face Transformers; bert-base-uncased; 768 features per comment
Numerical libraries	NumPy, pandas and SciPy for arrays, tabular input and numerical processing
Learning libraries	scikit-learn for ElasticNet; PyTorch for LSTM and GAN
Declared ML versions	torch 2.1.2, transformers 4.35.2, scikit-learn 1.3.2 and pandas 2.1.4 in requirements-ml.txt

Component	Specification and role
Development editor	Visual Studio Code or another Python-capable IDE can edit the project; the training log does not identify an IDE
Background execution	Python thread for the PANDORA experiment; Celery and Redis support other project workflows
Client software	A current browser with JavaScript enabled; Microsoft Word or compatible reader for exported reports

These package versions describe dependency declarations rather than a verified inventory of the training machine. The general requirements file pins NumPy 2.4.5, whereas the machine learning requirements file pins NumPy 1.26.3. They cannot both describe one installed version. Reproduction requires a tested, consistent environment and a saved package list. The experiment artifacts do not preserve the full installed versions, processor specification or GPU model.

Table 4.2: Practical starting specifications for local operation

Resource	Viewing results	Local model execution
Processor	64-bit dual-core processor	64-bit processor with at least four cores
Memory	4 GB RAM for the browser	16 GB RAM as a starting allocation; long sequences may require more
Storage	Space for downloads	At least 20 GB free for code, model downloads and a small working cache; data growth requires more
Operating system	Supported desktop OS	64-bit Windows or Linux with a compatible Python environment
GPU	Not required	Optional compatible CUDA GPU; CPU execution is possible but may be slow
Network	Access to the application	Required initially for packages and pretrained BERT downloads

Table 4.2 gives planning specifications, not experimentally certified minimum hardware. Resource needs depend on the number of authors, comment lengths, cached embeddings and the selected pipeline. In particular, the saved augmented baseline LSTM took more than fifteen hours in this session. A minimum configuration for the largest workload can only be established by measuring memory and runtime on that workload. The

browser requirements should therefore be distinguished from the server resources needed to train the models.

4.2 Implementation Architecture and Subsystem Modules

4.2.1 Client and server interaction

The system follows a client-server architecture implemented with Django's model-template-view organization. A browser sends an authenticated request to a URL route. The corresponding view validates the form, reads or creates database records and calls the required processing service. The template renders the resulting context as a page. Django models provide persistent records; the trained prediction models are separate machine learning objects loaded from saved artifacts.

A training request creates a PANDORA experiment record and starts a background Python thread. The experiment runner prepares the author partitions, selects comments, creates BERT representations and fits the eight conditions. When processing ends, the view stores condition results and artifact locations for later retrieval. This arrangement allows the researcher to return to a status page while processing continues, but the thread still depends on the application process remaining alive.

Table 4.3: Subsystems and information passed between them

Subsystem	Input and responsibility	Output or next component
Accounts and forms	Credentials and experiment or prediction settings	Authenticated, validated request
Dataset loader	Three prepared PANDORA workbooks	Author IDs, comment collections and five labels
Comment selector	Cleaned author comments; baseline or Q-learning policy	Selected comments for BERT
BERT encoder	Selected text, maximum 256 tokens per comment	768-dimensional comment vectors
Sparse branch	Mean vector for each author	ElasticNet estimates of five continuous traits
Sequence branch	Ordered comment-vector sequences	Two-layer bidirectional LSTM estimates
GAN augementer	Training embeddings and training scores only	Synthetic paired data for augmented fitting

Subsystem	Input and responsibility	Output or next component
Metrics and thresholds	True labels, scores and validation thresholds	Regression and binary classification measures
Persistence and interface	Runs, conditions, profiles and artifact paths	Experiment history, result tables and profile pages

The baseline training selector retains the available comment history. The Q-learning selector learns select-or-skip decisions from training comments using text informativeness, redundancy and a selection cost; its reward is not the final personality error. Selected text then reaches BERT. The sparse branch averages the comment vectors, while the LSTM branch retains a sequence. GAN augmentation is applied to training representations and labels only. It does not add independent validation or test authors.

The saved condition names use the prefix lasso, but the selected session explicitly configures ElasticNet with alpha 0.001 and l1_ratio 0.5. The chapter therefore calls this branch sparse regression or ElasticNet. The LSTM has two bidirectional recurrent layers, 128 hidden units per direction, dropout 0.2, a learning rate of 0.001, batch size four and a maximum of 35 epochs. The GAN configuration specifies 150 epochs, latent dimension 64, hidden dimension 128, batch size 16 and learning rate 0.0002. Each augmented condition records 50 generated rows. These are derived training examples, not additional people.

4.2.2 Dataset source and experimental allocation

The source is PANDORA, a Reddit comment dataset linked with personality and demographic labels. Gjurmović et al. (2021) describe approximately 10,000 users overall, including approximately 1,600 with Big Five labels. These publication-level totals describe the source corpus, not the number used in this project's selected experiment. The immediate project inputs are Training dataset.xlsx, Validation dataset.xlsx and Test dataset.xlsx in the PANDORA folder.

The selected session is database experiment 17, pandora_20260916_202608, with artifacts in run_20260917_133820. Its 50 training authors satisfy the requested upper limit of 200 training users. All eight conditions share 50 training, ten validation and ten test authors, with random seed 42. File-defined partitions determine membership; the actual allocation is

71.43%, 14.29% and 14.29% of the 70 sampled authors. It must not be described as a newly applied 60:20:20 split merely because ratio settings remain in the configuration.

Table 4.4: Authors and comments in the selected training session

Split	Authors	Comments	Mean per author	Median	Range
Training	50	8675	173.5	77.5	6–1401
Validation	10	681	68.1	57.0	6–163
Test	10	1663	166.3	23.5	10–1088
Total	70	11,019	157.4	—	—

The experiment contains 11,019 comments: 8,675 in training, 681 in validation and 1,663 in test. The saved overlap checks report zero common authors for every pair of partitions. Authors, rather than comments, are the independent labelled observations. Comment histories vary greatly, so the comment total does not imply 11,019 independent personality labels. The runner loaded the files as already cleaned and reports no additional comment removals during this run; this does not describe all earlier preparation of the source dataset.

The source scores are divided by 100 to obtain values on the zero-to-one scale. Binary ground truth uses training medians: 0.7900 for Openness, 0.3550 for Conscientiousness, 0.2500 for Extraversion, 0.3500 for Agreeableness and 0.4600 for Neuroticism. A score at or above its boundary is High. These ground-truth boundaries are different from the model-specific decision thresholds selected on validation data.

4.2.3 Storage and the saved prediction path

The principal database entities are experiment runs, condition results, threshold results, dataset allocations, prediction runs and test profiles. A prediction run links to its source experiment and selected condition. Each test profile retains the observed and predicted continuous traits, binary labels, decision thresholds and correctness flags. Consequently, the aggregate prediction measures can be recalculated from the stored profiles rather than accepted only from a dashboard summary.

The operational prediction helper embeds the first ten non-empty cleaned comments for every author. It does not reapply the training experiment's full-history baseline or learned Q-selection policy. Pipeline names in prediction records therefore identify the fitted model's training condition; they do not demonstrate identical comment selection at inference. This difference is material when interpreting a training-to-prediction change or attributing a gain specifically to Q-learning.

4.3 System Interface and Screen Shots

The interface separates authentication, experiment execution and prediction inspection. The following descriptions identify the principal implemented pages and the content required in their corresponding figures. Performance conclusions are based on saved research records rather than interface appearance.

4.3.1 Login and researcher dashboard

The login page controls access to the research functions through Django authentication. After authentication, the dashboard summarizes available records and recent PANDORA runs. It provides navigation to the tools used for training and prediction. Personal account details should be hidden in any reproduced interface image.

Figure 4.1: Login page and researcher dashboard — screenshot pending.

4.3.2 Experiment setup and results

The training interface accepts experiment settings and starts a PANDORA run. The experiment history lists saved runs, while the detail page identifies the run, its state and the eight condition results. A suitable screenshot should show experiment 17 and its completed results so that the interface evidence corresponds to the numerical tables in this chapter.

Figure 4.2: Training setup and completed eight-condition experiment — screenshot pending.

4.3.3 Prediction results and individual profiles

The prediction results page displays the batch identifier, fitted condition, number of profiles and summary classification metrics. Its displayed average decision threshold is only a summary of five thresholds; prediction uses a separate threshold for each trait. The profile

page compares observed and predicted OCEAN values and their Low or High labels. These views support inspection of individual errors as well as overall scores.

Figure 4.3: Saved prediction batch and anonymized OCEAN profile — screenshot pending.

4.4 Experiments and Results

4.4.1 Experimental conditions and evidence scope

The experiment varies prediction model, selection method and augmentation in a two-by-two-by-two design. E1–E8 below are short labels used consistently in the tables. They represent eight fitted conditions, not eight folds of cross-validation. The selected source experiment contains 70 real authors in total; each model is fitted with the same 50 real training authors.

Table 4.5: Definition of the eight experiments

ID	Saved condition	Predictor	Selection	GAN
E1	lasso_baseline	ElasticNet	Baseline	No
E2	lasso_baseline_gan	ElasticNet	Baseline	Yes
E3	lasso_qlearn	ElasticNet	Q-learning	No
E4	lasso_qlearn_gan	ElasticNet	Q-learning	Yes
E5	lstm_baseline	BiLSTM	Baseline	No
E6	lstm_baseline_gan	BiLSTM	Baseline	Yes
E7	lstm_qlearn	BiLSTM	Q-learning	No
E8	lstm_qlearn_gan	BiLSTM	Q-learning	Yes

The prediction comparison uses the latest eight successful prediction batches, ordered by creation time and record ID. The failed record 36 is excluded. The resulting records are 40, 39, 38, 37, 35, 34, 33 and 32. All link to experiment 17. They contain seven distinct pipelines because E6 occurs twice; E2 has no result within this requested window. E2 is retained in the training tables and marked unavailable in prediction comparisons rather than assigned a result from an older batch.

Seven batches contain the same 33 authors. Record 33 contains a 20-author subset. These batches reuse profiles and are not eight independent test samples. The 33-author set

has no overlap with the training or validation authors, but includes all ten original test authors. It therefore extends the observed test sample rather than providing a completely independent external test. Comparisons used to choose a pipeline are restricted to the equal 33-author batches.

4.4.2 Performance measures and reporting conventions

Mean absolute error (MAE) is the mean absolute difference between the observed and predicted continuous scores. Root mean squared error (RMSE) gives more weight to large errors. Lower MAE and RMSE indicate better predictions. R^2 compares squared error with prediction by the evaluated sample mean; a negative value indicates worse performance than that reference. Pearson correlation summarizes linear association and is reported as zero for constant predictions to match the project convention.

For binary evaluation, TP and TN denote correctly predicted High and Low labels, while FP and FN denote false High and false Low labels. Accuracy = $(TP + TN)/N$; precision = $TP/(TP + FP)$; recall = $TP/(TP + FN)$; specificity = $TN/(TN + FP)$; and High-class F1 = $2TP/(2TP + FP + FN)$. Undefined precision or F1 denominators are assigned zero, consistent with the saved prediction function. ROC AUC and PR AUC (average precision in the saved implementation) describe score ranking across operating points rather than performance at only the selected threshold.

Validation tables use an unweighted mean across the five traits. The application's saved prediction summary pools all author-trait binary decisions before calculating precision, recall, F1 and specificity. These pooled quantities differ from the mean of five trait metrics. The chapter presents both forms with explicit labels and uses trait means for training-to-prediction comparisons. RMSE is likewise averaged over the five trait-specific RMSE values; it is not a single square root after pooling all trait errors.

Some saved sparse-regression accuracy and macro-F1 fields refer to three classes, whereas the threshold fields refer to binary Low and High classes. Copying them into one binary comparison would mix definitions. The binary validation tables below use the retained binary precision, recall, F1 and specificity. Sparse binary accuracy is recovered from the saved positive and negative class counts, recall and specificity; the resulting confusion counts are integers. LSTM binary values come from the validation sweep row at the chosen

threshold. The test-stage `val_mae` field is identified by its containing test block and is reported as test MAE, not validation MAE.

4.4.3 Model fitting and overall validation performance

Table 4.6: Stored training-session performance and fitting cost

ID	Real train authors	Synthetic rows	Train MAE	Val MAE	Test MAE	Fit seconds
E1	50	0	0.2420	0.2569	0.2308	12.89
E2	50	50	0.1475	0.2620	0.2205	186.04
E3	50	0	0.2280	0.2618	0.2193	13.65
E4	50	50	0.1499	0.2495	0.2284	50.59
E5	50	0	NR	0.2470	0.2303	591.96
E6	50	50	NR	0.2503	0.2265	56253.64
E7	50	0	NR	0.2487	0.2292	256.48
E8	50	50	NR	0.2500	0.2296	883.46

NR = not recorded. Training MAE is the stored branch diagnostic; augmented diagnostics are not asserted to be measured on real authors alone. The source does not retain equivalent training-set metrics or epoch histories for the LSTM conditions. Validation and test values are explicitly separated from in-sample training performance.

The lowest validation MAE is E5 at 0.24698, while the lowest original ten-author test MAE is E3 at 0.21934. This difference shows that the preferred condition depends on the evaluated authors. The absence of saved LSTM training-set histories prevents a complete learning-curve or overfitting comparison. Training time is recorded by condition and is descriptive of this run; cache state and machine load were not controlled sufficiently to interpret it as a general hardware benchmark.

Table 4.7: All pipelines across validation regression and ranking metrics

ID	MAE	RMSE	R ²	Pearson r	ROC AUC	PR AUC
E1	0.2569	0.2999	-0.0781	0.0019	0.5417	0.6061
E2	0.2620	0.3070	-0.1248	-0.0030	0.5323	0.6610
E3	0.2618	0.3093	-0.1559	-0.0183	0.5167	0.5687
E4	0.2495	0.2948	-0.0399	0.1235	0.5567	0.6483

ID	MAE	RMSE	R ²	Pearson r	ROC AUC	PR AUC
E5	0.2470	0.2932	-0.0284	0.2000	0.4997	0.5963
E6	0.2503	0.2930	-0.0300	0.1515	0.4823	0.5915
E7	0.2487	0.2918	-0.0185	0.0629	0.3913	0.5430
E8	0.2500	0.2943	-0.0398	0.2357	0.5793	0.6511

Table 4.8: All pipelines across consistent binary validation metrics

ID	Accuracy	Precision	Recall	F1 High	Specificity
E1	0.6000	0.5200	0.3500	0.3633	0.8000
E2	0.6400	0.8033	0.4567	0.5556	0.8300
E3	0.5600	0.3867	0.3500	0.3405	0.7333
E4	0.6600	0.7124	0.6900	0.6803	0.6467
E5	0.6400	0.6383	0.7100	0.6581	0.5833
E6	0.6200	0.6183	0.6433	0.6172	0.5900
E7	0.5400	0.5583	0.5033	0.5061	0.5900
E8	0.7000	0.7050	0.7333	0.6983	0.6800

Each validation trait is represented by only ten authors, so moving one classification changes its accuracy by 0.10. Threshold tuning and reporting use the same validation authors. Validation classification scores are consequently development results, not unbiased estimates from an untouched sample. The separate ten-author test and later saved predictions must be considered alongside them.

4.4.4 Validation performance across the five traits

Table 4.9: Per-trait training diagnostics and validation regression performance

ID	Trait	Train MAE	Val MAE	Val RMSE	Val R ²	Val r
E1	O	0.1821	0.2130	0.2394	-0.0136	0.5206
E1	C	0.2837	0.2536	0.2982	-0.0011	0.0000
E1	E	0.2779	0.2785	0.3229	-0.0130	0.0000
E1	A	0.2214	0.2473	0.2873	-0.0014	0.0000

ID	Trait	Train MAE	Val MAE	Val RMSE	Val R ²	Val r
E1	N	0.2450	0.2919	0.3516	-0.3613	-0.5113
E2	O	0.1163	0.2078	0.2305	0.0601	0.5091
E2	C	0.1590	0.2793	0.3264	-0.1989	-0.2031
E2	E	0.1699	0.3067	0.3492	-0.1843	-0.2029
E2	A	0.1292	0.2237	0.2725	0.0992	0.3753
E2	N	0.1631	0.2926	0.3566	-0.4000	-0.4935
E3	O	0.1902	0.2042	0.2367	0.0090	0.3607
E3	C	0.2837	0.2536	0.2982	-0.0011	0.0000
E3	E	0.2779	0.2785	0.3229	-0.0130	0.0000
E3	A	0.2214	0.2473	0.2873	-0.0014	0.0000
E3	N	0.1667	0.3252	0.4013	-0.7729	-0.4521
E4	O	0.1220	0.1823	0.2221	0.1275	0.5320
E4	C	0.1607	0.2772	0.3296	-0.2232	-0.1777
E4	E	0.1705	0.2804	0.3247	-0.0239	-0.0481
E4	A	0.1501	0.2149	0.2554	0.2086	0.6192
E4	N	0.1461	0.2929	0.3421	-0.2886	-0.3077
E5	O	NR	0.1888	0.2360	0.0147	0.2790
E5	C	NR	0.2575	0.3037	-0.0384	-0.1322
E5	E	NR	0.2740	0.3183	0.0159	0.2761
E5	A	NR	0.2432	0.2834	0.0255	0.6044
E5	N	NR	0.2714	0.3245	-0.1595	-0.0272
E6	O	NR	0.2120	0.2459	-0.0692	0.1002
E6	C	NR	0.2527	0.2983	-0.0018	0.0620
E6	E	NR	0.2763	0.3205	0.0022	0.2482
E6	A	NR	0.2463	0.2853	0.0123	0.7389
E6	N	NR	0.2640	0.3152	-0.0935	-0.3917
E7	O	NR	0.1950	0.2378	-0.0004	0.0484
E7	C	NR	0.2584	0.3042	-0.0413	-0.3098
E7	E	NR	0.2781	0.3211	-0.0015	0.3478

ID	Trait	Train MAE	Val MAE	Val RMSE	Val R ²	Val r
E7	A	NR	0.2449	0.2841	0.0204	0.6771
E7	N	NR	0.2671	0.3117	-0.0699	-0.4489
E8	O	NR	0.2050	0.2425	-0.0403	0.1936
E8	C	NR	0.2527	0.2983	-0.0014	0.1073
E8	E	NR	0.2713	0.3152	0.0352	0.4198
E8	A	NR	0.2394	0.2795	0.0520	0.8621
E8	N	NR	0.2818	0.3362	-0.2443	-0.4041

O = Openness; C = Conscientiousness; E = Extraversion; A = Agreeableness; N = Neuroticism. All validation rows use ten authors. NR indicates an absent stored training metric, not zero error.

The trait breakdown prevents an overall mean from hiding uneven performance. For example, E1 has validation MAE 0.2130 for Openness and 0.2919 for Neuroticism. Its constant predictions on several other traits have Pearson correlation recorded as zero. Such results indicate that a pipeline may learn some variation in one target while effectively predicting a common score for another. Claims about the entire OCEAN profile therefore require all five traits, not only the best-performing trait.

4.4.5 Selection of a decision threshold for each trait

Threshold selection begins after continuous prediction. The training-median boundary creates the observed High or Low label, while a candidate decision threshold converts the model's validation score into a predicted label. With ten validation scores per trait, the implementation uses the distinct predicted score values, clipped to the zero-to-one interval and rounded to four decimal places, as candidates. The general implementation uses percentiles when there are more than nineteen scores. The fixed fallback list in the configuration is therefore not the complete set of thresholds actually considered in this session.

For each candidate, the selector computes binary metrics and maximizes the harmonic mean of High-class F1 and specificity: $H = 2 \times F1 \times \text{specificity} / (F1 + \text{specificity})$. H is zero when its denominator is zero. The intent is to discourage selecting High for every author simply to obtain a large recall. Ties are resolved by higher F1, then higher specificity, then

higher accuracy; a remaining exact tie retains the first candidate in the sorted list. The selected validation threshold is frozen for test evaluation and loaded by the prediction function.

Table 4.10: Selected trait thresholds and binary validation performance at each threshold

ID	Trait	Threshold	Accuracy	Precision	Recall	F1	Specificity
E1	O	0.6781	0.7000	1.0000	0.5000	0.6667	1.0000
E1	C	0.4395	0.6000	0.6000	1.0000	0.7500	0.0000
E1	E	0.3776	0.5000	0.0000	0.0000	0.0000	1.0000
E1	A	0.4117	0.5000	0.0000	0.0000	0.0000	1.0000
E1	N	0.4814	0.7000	1.0000	0.2500	0.4000	1.0000
E2	O	0.6845	0.7000	1.0000	0.5000	0.6667	1.0000
E2	C	0.4835	0.5000	0.6667	0.3333	0.4444	0.7500
E2	E	0.3636	0.6000	0.6000	0.6000	0.6000	0.6000
E2	A	0.4393	0.7000	0.7500	0.6000	0.6667	0.8000
E2	N	0.4685	0.7000	1.0000	0.2500	0.4000	1.0000
E3	O	0.6883	0.7000	1.0000	0.5000	0.6667	1.0000
E3	C	0.4395	0.6000	0.6000	1.0000	0.7500	0.0000
E3	E	0.3776	0.5000	0.0000	0.0000	0.0000	1.0000
E3	A	0.4117	0.5000	0.0000	0.0000	0.0000	1.0000
E3	N	0.4754	0.5000	0.3333	0.2500	0.2857	0.6667
E4	O	0.6906	0.8000	0.8333	0.8333	0.8333	0.7500
E4	C	0.4450	0.7000	0.8000	0.6667	0.7273	0.7500
E4	E	0.3443	0.5000	0.5000	0.6000	0.5455	0.4000
E4	A	0.4522	0.8000	1.0000	0.6000	0.7500	1.0000
E4	N	0.3558	0.5000	0.4286	0.7500	0.5455	0.3333
E5	O	0.7551	0.7000	0.8000	0.6667	0.7273	0.7500
E5	C	0.4865	0.4000	0.5000	0.3333	0.4000	0.5000
E5	E	0.3479	0.7000	0.6250	1.0000	0.7692	0.4000
E5	A	0.3911	0.7000	0.6667	0.8000	0.7273	0.6000

ID	Trait	Threshold	Accuracy	Precision	Recall	F1	Specificity
E5	N	0.4667	0.7000	0.6000	0.7500	0.6667	0.6667
E6	O	0.6904	0.6000	0.7500	0.5000	0.6000	0.7500
E6	C	0.4437	0.6000	0.6667	0.6667	0.6667	0.5000
E6	E	0.3231	0.7000	0.6250	1.0000	0.7692	0.4000
E6	A	0.4141	0.8000	0.8000	0.8000	0.8000	0.8000
E6	N	0.4352	0.4000	0.2500	0.2500	0.2500	0.5000
E7	O	0.7393	0.5000	0.6667	0.3333	0.4444	0.7500
E7	C	0.4860	0.4000	0.5000	0.3333	0.4000	0.5000
E7	E	0.3821	0.7000	0.6250	1.0000	0.7692	0.4000
E7	A	0.4104	0.7000	0.7500	0.6000	0.6667	0.8000
E7	N	0.3991	0.4000	0.2500	0.2500	0.2500	0.5000
E8	O	0.6837	0.8000	0.8333	0.8333	0.8333	0.7500
E8	C	0.4523	0.5000	0.6667	0.3333	0.4444	0.7500
E8	E	0.3491	0.7000	0.6250	1.0000	0.7692	0.4000
E8	A	0.3980	1.0000	1.0000	1.0000	1.0000	1.0000
E8	N	0.4696	0.5000	0.4000	0.5000	0.4444	0.5000

A selected threshold is not proof of useful separation. E1, for example, records zero High-class F1 for Extraversion and Agreeableness despite specificity of 1.0000. Those decisions classify every validation example as Low. The threshold is an operational choice from the observed scores; it is not a population personality boundary or a calibrated probability cutoff. Test-only threshold sweeps are descriptive and are not used to replace the validation-selected values.

4.4.6 The latest eight successful prediction batches

Table 4.11: Prediction-batch inventory and continuous performance

Record	ID	Time UTC	Requested	Predicted	Source comments	MAE
40	E7	14:55:59	200	33	3936	0.2484
39	E5	14:48:35	200	33	3936	0.2469

Record	ID	Time UTC	Requested	Predicted	Source comments	MAE
38	E8	14:41:58	200	33	3936	0.2466
37	E3	14:35:00	185	33	3936	0.2540
35	E1	14:30:42	200	33	3936	0.2540
34	E6	14:26:54	200	33	3936	0.2472
33	E6	14:25:25	20	20	1639	0.2441
32	E4	14:15:20	33	33	3936	0.2676

All batches were created on 17 September 2026 and refer to experiment 17. Times are the stored UTC timestamps. Source comments count available comments attached to profiles, not comments embedded: the operational helper processes at most ten per author. Requested sample size does not equal the number available for prediction.

Table 4.12: Each saved prediction batch across the application classification metrics

Record / ID	Accuracy	Precision	Recall	F1 High	Specificity
40 / E7	0.4606	0.4604	0.8205	0.5899	0.1379
39 / E5	0.4667	0.4561	0.6667	0.5417	0.2874
38 / E8	0.4667	0.4603	0.7436	0.5686	0.2184
37 / E3	0.4970	0.4615	0.3846	0.4196	0.5977
35 / E1	0.4909	0.4531	0.3718	0.4085	0.5977
34 / E6	0.4667	0.4615	0.7692	0.5769	0.1954
33 / E6	0.4700	0.4750	0.7755	0.5891	0.1765
32 / E4	0.4788	0.4608	0.6026	0.5222	0.3678

These values are independently checked against the stored binary profile labels. Metrics pool five decisions per author: 165 decisions in each 33-author batch and 100 in the 20-author batch. They are not eight repetitions of a common independent experiment.

Table 4.13: Each saved prediction batch across continuous and ranking metrics

Record / ID	n	MAE	RMSE	R ²	r	ROC AUC	PR AUC
40 / E7	33	0.2484	0.2928	-0.0517	0.0357	0.5160	0.5450
39 / E5	33	0.2469	0.2906	-0.0353	0.0463	0.5308	0.5544
38 / E8	33	0.2466	0.2873	-0.0119	0.0629	0.5309	0.5526
37 / E3	33	0.2540	0.3015	-0.1148	-0.0015	0.5044	0.5108

Record / ID	n	MAE	RMSE	R ²	r	ROC AUC	PR AUC
35 / E1	33	0.2540	0.2949	-0.0672	0.0026	0.4978	0.4994
34 / E6	33	0.2472	0.2879	-0.0145	0.0461	0.5212	0.5504
33 / E6	20	0.2441	0.2861	-0.0379	0.0340	0.4828	0.5359
32 / E4	33	0.2676	0.3146	-0.2130	-0.0647	0.4799	0.5118

Prediction ROC AUC and average precision are recalculated from the saved continuous scores and binary ground truth, with ties handled consistently. The equal-size comparison gives E8 the lowest prediction MAE and RMSE, while E3 gives the highest pooled binary accuracy. E7 gives the highest pooled F1 and recall, but its specificity is only 0.1379, showing that it often predicts High for truly Low cases. Record 33 has a smaller MAE than the 33-author batches, but this cannot establish E6 as the best pipeline because it uses a different 20-author subset. Both E6 records remain visible to preserve the requested latest-eight selection.

4.4.7 Training and validation compared with actual prediction

Table 4.14: Direct validation and prediction comparison using trait-mean regression metrics

ID	Stage	MAE	RMSE	R ²	r	ROC AUC	PR AUC
E1	Validation	0.2569	0.2999	-0.0781	0.0019	0.5417	0.6061
E1	Prediction 35	0.2540	0.2949	-0.0672	0.0026	0.4978	0.4994
E2	Validation	0.2620	0.3070	-0.1248	-0.0030	0.5323	0.6610
E2	Not in window	—	—	—	—	—	—
E3	Validation	0.2618	0.3093	-0.1559	-0.0183	0.5167	0.5687
E3	Prediction 37	0.2540	0.3015	-0.1148	-0.0015	0.5044	0.5108
E4	Validation	0.2495	0.2948	-0.0399	0.1235	0.5567	0.6483
E4	Prediction 32	0.2676	0.3146	-0.2130	-0.0647	0.4799	0.5118
E5	Validation	0.2470	0.2932	-0.0284	0.2000	0.4997	0.5963
E5	Prediction 39	0.2469	0.2906	-0.0353	0.0463	0.5308	0.5544
E6	Validation	0.2503	0.2930	-0.0300	0.1515	0.4823	0.5915
E6	Prediction 34	0.2472	0.2879	-0.0145	0.0461	0.5212	0.5504

ID	Stage	MAE	RMSE	R ²	r	ROC AUC	PR AUC
E7	Validation	0.2487	0.2918	-0.0185	0.0629	0.3913	0.5430
E7	Prediction 40	0.2484	0.2928	-0.0517	0.0357	0.5160	0.5450
E8	Validation	0.2500	0.2943	-0.0398	0.2357	0.5793	0.6511
E8	Prediction 38	0.2466	0.2873	-0.0119	0.0629	0.5309	0.5526

Table 4.15: Direct validation and prediction comparison using trait-mean binary metrics

ID	Stage	Accuracy	Precision	Recall	F1	Specificity
E1	Validation	0.6000	0.5200	0.3500	0.3633	0.8000
E1	Prediction 35	0.4909	0.2707	0.3800	0.3047	0.6052
E2	Validation	0.6400	0.8033	0.4567	0.5556	0.8300
E2	Not in window	—	—	—	—	—
E3	Validation	0.5600	0.3867	0.3500	0.3405	0.7333
E3	Prediction 37	0.4970	0.2784	0.3933	0.3153	0.6059
E4	Validation	0.6600	0.7124	0.6900	0.6803	0.6467
E4	Prediction 32	0.4788	0.4608	0.5954	0.5113	0.3653
E5	Validation	0.6400	0.6383	0.7100	0.6581	0.5833
E5	Prediction 39	0.4667	0.4355	0.6770	0.5013	0.2970
E6	Validation	0.6200	0.6183	0.6433	0.6172	0.5900
E6	Prediction 34	0.4667	0.3691	0.7867	0.5024	0.2111
E7	Validation	0.5400	0.5583	0.5033	0.5061	0.5900
E7	Prediction 40	0.4606	0.4544	0.8196	0.5760	0.1389
E8	Validation	0.7000	0.7050	0.7333	0.6983	0.6800
E8	Prediction 38	0.4667	0.3677	0.7600	0.4949	0.2333

Prediction rows use the newest batch for each pipeline inside the latest-eight window, all with 33 authors. Record 33 is still reported in Tables 4.11–4.13 but is not substituted for the larger E6 comparison. E2 is unavailable in this window. Binary prediction F1 here is a mean over traits and differs from the pooled F1 in Table 4.12.

These paired tables align metric definitions, but they do not make the evaluated populations or comment-selection procedures identical. Validation uses ten authors and the

experimental feature path. Prediction uses 33 authors and the first-ten-comment helper. Changes therefore combine sample composition, threshold behaviour and an implementation difference. They cannot be attributed solely to overfitting, Q-learning or augmentation.

4.4.8 Detailed outcomes of the eight experiments

E1 ElasticNet with baseline selection

E1 is the pooled-BERT sparse baseline. It establishes the effect of fitting ElasticNet without learned selection or augmentation. Several targets have no active sparse features, which limits their ability to follow author-level variation.

Validation MAE was 0.2569, RMSE 0.2999, R^2 -0.0781 and Pearson r 0.0019. Mean binary validation accuracy was 0.6000, precision 0.5200, recall 0.3500, F1 0.3633 and specificity 0.8000. The original test MAE was 0.2308, compared with a recorded fitting time of 12.89 seconds. Trait-level errors and threshold-specific metrics are given in Tables 4.9 and 4.10.

In prediction record 35 (33 authors), MAE was 0.2540, RMSE 0.2949, mean R^2 -0.0672 and mean Pearson r 0.0026. The application's pooled accuracy was 0.4909, precision 0.4531, recall 0.3718, F1 0.4085 and specificity 0.5977. Tables 4.14 and 4.15 give the corresponding validation-to-prediction comparison under common trait-mean definitions.

E2 ElasticNet with baseline selection and GAN

E2 adds training-only GAN augmentation to E1. Its original test MAE is lower than E1, but its validation MAE is higher. This mixed result does not support a general claim that augmentation improves sparse prediction. There is no E2 batch in the latest eight successful predictions, so its operational performance is not ranked.

Validation MAE was 0.2620, RMSE 0.3070, R^2 -0.1248 and Pearson r -0.0030. Mean binary validation accuracy was 0.6400, precision 0.8033, recall 0.4567, F1 0.5556 and specificity 0.8300. The original test MAE was 0.2205, compared with a recorded fitting time of 186.04 seconds. Trait-level errors and threshold-specific metrics are given in Tables 4.9 and 4.10.

E3 ElasticNet with Q learning selection

E3 uses Q-learning-selected BERT comments with ElasticNet and no GAN. It has the lowest original ten-author test MAE. In actual prediction it achieves the best pooled accuracy among the equally sized batches, but its low recall shows that this accuracy does not imply equally good identification of High labels.

Validation MAE was 0.2618, RMSE 0.3093, R^2 -0.1559 and Pearson r -0.0183. Mean binary validation accuracy was 0.5600, precision 0.3867, recall 0.3500, F1 0.3405 and specificity 0.7333. The original test MAE was 0.2193, compared with a recorded fitting time of 13.65 seconds. Trait-level errors and threshold-specific metrics are given in Tables 4.9 and 4.10.

In prediction record 37 (33 authors), MAE was 0.2540, RMSE 0.3015, mean R^2 -0.1148 and mean Pearson r -0.0015. The application's pooled accuracy was 0.4970, precision 0.4615, recall 0.3846, F1 0.4196 and specificity 0.5977. Tables 4.14 and 4.15 give the corresponding validation-to-prediction comparison under common trait-mean definitions.

E4 ElasticNet with Q learning selection and GAN

E4 combines sparse regression, Q-learning and GAN augmentation. Its validation MAE is better than E3, but its original test error and saved prediction error are worse. The combined additions therefore do not provide a consistent improvement over the simpler sparse condition.

Validation MAE was 0.2495, RMSE 0.2948, R^2 -0.0399 and Pearson r 0.1235. Mean binary validation accuracy was 0.6600, precision 0.7124, recall 0.6900, F1 0.6803 and specificity 0.6467. The original test MAE was 0.2284, compared with a recorded fitting time of 50.59 seconds. Trait-level errors and threshold-specific metrics are given in Tables 4.9 and 4.10.

In prediction record 32 (33 authors), MAE was 0.2676, RMSE 0.3146, mean R^2 -0.2130 and mean Pearson r -0.0647. The application's pooled accuracy was 0.4788, precision 0.4608, recall 0.6026, F1 0.5222 and specificity 0.3678. Tables 4.14 and 4.15 give the corresponding validation-to-prediction comparison under common trait-mean definitions.

E5 Bidirectional LSTM with baseline selection

E5 replaces pooled regression with a bidirectional sequence model and uses the baseline comment history during training. It gives the lowest validation MAE. Its prediction error is close to E8, while its pooled F1 and recall are lower and its specificity is higher.

Validation MAE was 0.2470, RMSE 0.2932, R^2 -0.0284 and Pearson r 0.2000. Mean binary validation accuracy was 0.6400, precision 0.6383, recall 0.7100, F1 0.6581 and specificity 0.5833. The original test MAE was 0.2303, compared with a recorded fitting time of 591.96 seconds. Trait-level errors and threshold-specific metrics are given in Tables 4.9 and 4.10.

In prediction record 39 (33 authors), MAE was 0.2469, RMSE 0.2906, mean R^2 -0.0353 and mean Pearson r 0.0463. The application's pooled accuracy was 0.4667, precision 0.4561, recall 0.6667, F1 0.5417 and specificity 0.2874. Tables 4.14 and 4.15 give the corresponding validation-to-prediction comparison under common trait-mean definitions.

E6 Bidirectional LSTM with baseline selection and GAN

E6 adds GAN augmentation to the baseline LSTM. Its original test RMSE is slightly lower than the other conditions, but its recorded fitting duration is exceptionally long. The two saved prediction batches demonstrate why results from 20 and 33 authors must not be treated as matched repetitions.

Validation MAE was 0.2503, RMSE 0.2930, R^2 -0.0300 and Pearson r 0.1515. Mean binary validation accuracy was 0.6200, precision 0.6183, recall 0.6433, F1 0.6172 and specificity 0.5900. The original test MAE was 0.2265, compared with a recorded fitting time of 56253.64 seconds. Trait-level errors and threshold-specific metrics are given in Tables 4.9 and 4.10.

In prediction record 34 (33 authors), MAE was 0.2472, RMSE 0.2879, mean R^2 -0.0145 and mean Pearson r 0.0461. The application's pooled accuracy was 0.4667, precision 0.4615, recall 0.7692, F1 0.5769 and specificity 0.1954. Tables 4.14 and 4.15 give the corresponding validation-to-prediction comparison under common trait-mean definitions.

E7 Bidirectional LSTM with Q learning selection

E7 uses the sequence model trained on Q-selected comments without augmentation. Its prediction F1 and recall lead the equal-size saved batches, but low specificity reveals a pronounced tendency to predict High. This is a classification trade-off rather than a general improvement in all measures.

Validation MAE was 0.2487, RMSE 0.2918, R^2 -0.0185 and Pearson r 0.0629. Mean binary validation accuracy was 0.5400, precision 0.5583, recall 0.5033, F1 0.5061 and specificity 0.5900. The original test MAE was 0.2292, compared with a recorded fitting time of 256.48 seconds. Trait-level errors and threshold-specific metrics are given in Tables 4.9 and 4.10.

In prediction record 40 (33 authors), MAE was 0.2484, RMSE 0.2928, mean R^2 -0.0517 and mean Pearson r 0.0357. The application's pooled accuracy was 0.4606, precision 0.4604, recall 0.8205, F1 0.5899 and specificity 0.1379. Tables 4.14 and 4.15 give the corresponding validation-to-prediction comparison under common trait-mean definitions.

E8 Bidirectional LSTM with Q learning selection and GAN

E8 combines Q-learning, BERT, GAN augmentation and the bidirectional LSTM. It gives the lowest MAE and RMSE on the common 33-author prediction set. This supports its selection for continuous OCEAN estimation within the observed comparison, while the negative mean R^2 and low specificity limit the strength and scope of the recommendation.

Validation MAE was 0.2500, RMSE 0.2943, R^2 -0.0398 and Pearson r 0.2357. Mean binary validation accuracy was 0.7000, precision 0.7050, recall 0.7333, F1 0.6983 and specificity 0.6800. The original test MAE was 0.2296, compared with a recorded fitting time of 883.46 seconds. Trait-level errors and threshold-specific metrics are given in Tables 4.9 and 4.10.

In prediction record 38 (33 authors), MAE was 0.2466, RMSE 0.2873, mean R^2 -0.0119 and mean Pearson r 0.0629. The application's pooled accuracy was 0.4667, precision 0.4603, recall 0.7436, F1 0.5686 and specificity 0.2184. Tables 4.14 and 4.15 give the corresponding validation-to-prediction comparison under common trait-mean definitions.

4.4.9 Preferred pipeline during prediction

E8, the model trained with Q-learning selection, BERT representations, GAN augmentation and a bidirectional LSTM, is the preferred pipeline for continuous OCEAN estimation among the seven pipelines represented on the common 33-author prediction set. The selection rule prioritizes mean MAE, then mean RMSE. E8 records MAE 0.2466 and RMSE 0.2873, both the lowest in that matched comparison. This choice follows the primary continuous-score objective established in Chapters One and Three. It is not an assertion that E8 wins every classification metric or that the unrepresented E2 has been ruled out.

E8 improves MAE over E5 by 0.000345 and over E6 by 0.000560 on the same authors. Its improvement over E3 is 0.007356. These differences are descriptive: the small E8–E5 margin is not evidence of statistical significance. No repeated-seed analysis or confidence interval establishes a stable population ranking. All compared mean R^2 values remain negative, including E8 at -0.0119, so the selected pipeline still has weak continuous predictive performance.

Table 4.16: Selected E8 thresholds and actual per-trait prediction results

Trait	Label cutoff	Decision threshold	MAE	Accuracy	F1	Specificity
O	0.7900	0.6837	0.1964	0.4545	0.6250	0.0000
C	0.3550	0.4523	0.2799	0.4545	0.5714	0.1667
E	0.2500	0.3491	0.2066	0.4848	0.0000	1.0000
A	0.3500	0.3980	0.2792	0.4545	0.6250	0.0000
N	0.4600	0.4696	0.2710	0.4848	0.6531	0.0000

The operational thresholds for E8 are 0.6837 for Openness, 0.4523 for Conscientiousness, 0.3491 for Extraversion, 0.3980 for Agreeableness and 0.4696 for Neuroticism. They match the thresholds stored in its prediction profiles and originate from validation. Their arithmetic mean, 0.47054, is the value displayed at batch level; it should not replace the five thresholds in individual trait decisions.

Table 4.17: Full E8 per-trait prediction metrics at the selected operating thresholds

Trait	RMSE	R ²	Precision	Recall	TP / FP	TN / FN
O	0.2576	-0.0439	0.4545	1.0000	15 / 18	0 / 0
C	0.3189	-0.0866	0.4444	0.8000	12 / 15	3 / 3
E	0.2483	-0.0016	0.0000	0.0000	0 / 0	16 / 17
A	0.3150	0.0141	0.4545	1.0000	15 / 18	0 / 0
N	0.2965	0.0583	0.4848	1.0000	16 / 17	0 / 0

Across all 165 author-trait decisions, E8 produces 58 true High predictions, 19 true Low predictions, 68 false High predictions and 20 false Low predictions. Its pooled accuracy is 0.4667, F1 is 0.5686 and specificity is 0.2184. A classifier that always predicts Low on this set would obtain $87/165 = 0.5273$ accuracy but zero High-class recall and F1. Consequently, the selected continuous predictor cannot also be claimed to provide strong binary classification. E3 is preferable if the narrow criterion is observed binary accuracy among these pipelines, and E7 is preferable if the criterion is observed High-class F1; neither conclusion establishes general deployment quality.

4.4.10 Discussion of component effects and study limitations

Q-learning reduced the experiment's selected comment total from 11,019 to 7,598, a reduction of 3,421 comments or 31.05%. The average decreased from 157.41 to 108.54 comments per author. This is evidence of lower input volume during the experimental feature path, not a measured inference-time saving in the first-ten-comment helper. The recorded training times also vary substantially between conditions and include an unusually large 56,253.64 seconds for E6.

On the original test set, adding Q-learning without GAN reduced MAE for both the sparse and sequence branches, while adding it with GAN increased MAE for both branches. Augmentation likewise helped the baseline-selection test conditions but harmed the Q-learning test conditions. The directions differ again in the later prediction comparison. These interactions show why the combined pipeline must be evaluated directly instead of assuming that two potentially useful components will necessarily improve one another.

The strongest limitations are the 50-author training sample, ten-author validation set, one recorded random seed and reused test authors. Thresholds optimized on ten validation scores can be sensitive to individual authors. The saved source repository was marked dirty, so its commit identifier does not fully identify all code executed. Training histories and the full hardware inventory were not retained. Most importantly, operational inference uses a different comment-selection procedure from the source experiment. These constraints limit causal attribution and reproducibility, even though the saved prediction summaries can be checked against the stored profile labels.

The results support the implementation of an integrated eight-condition research system and a provisional selection of E8 for continuous prediction within the observed equal-size batches. They do not establish a universally best pipeline, reliable population-level personality assessment or a complete eight-pipeline operational comparison. Further evaluation should preserve raw predictions, reproduce each condition's selection path at inference, include E2 on the same held-out authors, and compare multiple seeds on a larger independent sample.

References

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Project evidence: SQLite experiment record 17 and successful prediction records 40, 39, 38, 37, 35, 34, 33 and 32; `run_20260917_133820/experiment_data.json`, `data_quality.json`, `imbalance_report.json`, `trait_label_thresholds.json`, `reproducibility.json`, `run_summary.json` and the eight condition `metrics.json` files. Results inspected on 18 September 2026. Implementation references: `backend/tools/views.py`, `backend/core/models.py`, `backend/ml_pipeline/experiments/pandora_runner.py` and `backend/ml_pipeline/services/metrics_engine.py`.